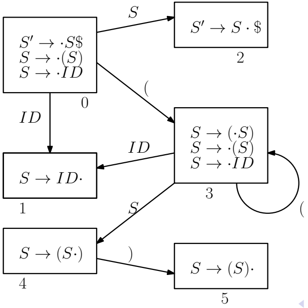
Conflicts in LR Parser
Compiler Design (CS 3007)

Sumanta Pyne

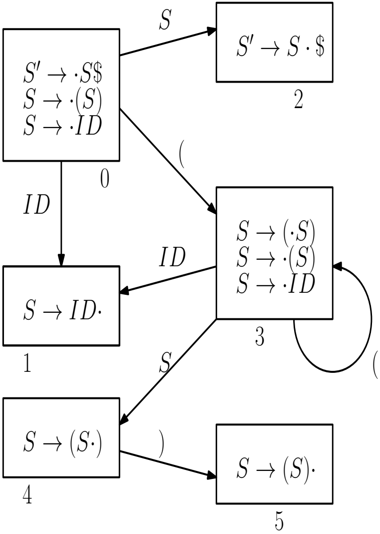
Assistant Professor
Computer Science and Engineering Department
National Institute of Technology Rourkela
pynes@nitrkl.ac.in

April 14, 2026

- Given a grammar with rules $S \rightarrow (S)|ID$
- Is the given grammar LR(0) ?
- Augmented grammar
  Rule 0:$S' \rightarrow S\$$
  Rule 1:$S \rightarrow (S)$
  Rule 2:$S \rightarrow ID$
- DFA of LR(0) set of items using GOTO and CLOSURE

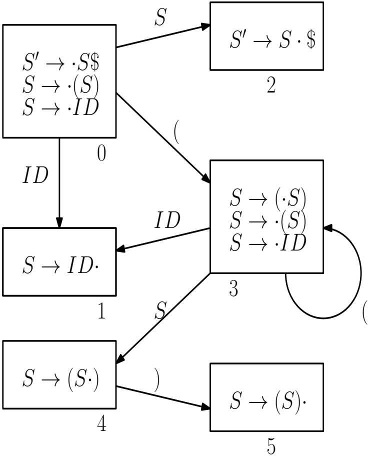


Construction of an LR(0) Parser



Construction of an LR(0) Parser

- Parsing table formation



	Action				Goto
State	(	)	\$	ID	S
0	$s3$			$s1$	2
1	$r2$	$r2$	r2	$r2$	
2			accept		
3	$s3$			$s1$	4
4		$s5$			
5	$r1$	$r1$	$r1$	$r1$	

- No conflicting entries
- Given grammar is LR(0)

Shift-Reduce conflict in LR(0) parsers

- General format of an LR(0) set of item with SR conflict
- Check SR conflict for LR(0) parsers

I_x

$p : A \rightarrow \alpha \cdot a\beta$

$q : B \rightarrow \gamma \cdot$

ACTION

	a	
	sy	
x	rq	

sy - shift a and goto state y

rq - reduce by rule q

$\text{ACTION}[x][a] = sy \qquad \text{ACTION}[x][a] = rq \forall a \in \Sigma \cup \{\$\}$

Shift-Reduce conflict in SLR(1) parsers

- General format of an LR(0) set of item with SR conflict
- Check SR conflict for SLR(1) parsers

I_x

$p : A \rightarrow \alpha \cdot a\beta$

$q : B \rightarrow \gamma \cdot$

ACTION

	a	
	sy	
x	rq	

sy - shift a and goto state y

rq - reduce by rule q

$\text{ACTION}[x][a] = sy \quad \text{ACTION}[x][a] = rq \forall a \in \text{FOLLOW}(B)$

Shift-Reduce conflict in LR(1) parsers

- General format of an LR(1) set of item with SR conflict
- Check SR conflict for CLR(1) and LALR(1) parsers

I_x

$p : A \rightarrow \alpha \cdot a\beta, b$

$q : B \rightarrow \gamma \cdot, a$

ACTION

	a
	sy
x	rq

sy - shift a and goto state y

rq - reduce by rule q

$\text{ACTION}[x][a] = sy \qquad \text{ACTION}[x][a] = rq$

Reduce-Reduce conflict in LR(1) parsers

- Two or more LR(1) set of items of CLR(1) parser having
  - different reduction rules
  - common lookahead symbols

I_x

$p : A \to \alpha\cdot, b$

I_y

$q : B \to \gamma\cdot, b$

ACTION

	b	
x	rp	
y	rq	

rp - reduce by rule p
rq - reduce by rule q

$\text{ACTION}[x][b] = rp$ $\quad$ $\text{ACTION}[y][b] = rq$

Reduce-Reduce conflict in LR(1) parsers

- General format of an LR(1) set of item with RR conflict
- Formed my merging two or more LR(1) set of items in CLR(1) parser
- Check RR conflict for LALR(1) parsers

I_{xy}
$p : A \rightarrow \alpha\cdot, b$
$q : B \rightarrow \gamma\cdot, b$

$\text{FOLLOW}(A) \cap \text{FOLLOW}(B) \neq \emptyset$

ACTION

	b
xy	rp, rq

rp - reduce by rule p
rq - reduce by rule q

$\text{ACTION}[xy][b] = rp \qquad \text{ACTION}[xy][b] = rq$

An SLR(1) grammar that is not LR(0)

$Rule\ 0 : E \rightarrow E + T$
$Rule\ 1 : E \rightarrow T$
$Rule\ 2 : T \rightarrow T * F$

$$
\begin{tikzcd}
\boxed{\begin{array}{l} E \rightarrow \cdot E + T \\ E \rightarrow \cdot T \\ T \rightarrow \cdot T * F \end{array}} \arrow[r, "T"] & \boxed{\begin{array}{l} E \rightarrow T \cdot \\ T \rightarrow T \cdot * F \end{array}} \arrow[r, "*"] & \boxed{T \rightarrow T * \cdot F} \\
0 & 1 & 2
\end{tikzcd}
$$

LR(0) table

State	+	*
1	$r1$	r1/s2

SR - conflict

If $A \rightarrow \alpha \cdot \in I_i$ then
ACTION$[i][x]=r''A \rightarrow \alpha'' \forall x \in \Sigma \cup \{\$\}$

SLR(1) table

State	+	*
1	$r1$	s2

If $A \rightarrow \alpha \cdot \in I_i$ then
ACTION$[i][x]=r''A \rightarrow \alpha'' \forall x \in$ FOLLOW(A)

Here $E \rightarrow T \cdot \in I_1$, $\{+\} \in$ FOLLOW(E)
ACTION$[1][+]=r1$

Thank you